# $C^3$: Committed Confidence Calibration Selection Bias in Confidence-Based Token Commitment in Diffusion Language Models

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## Abstract

Masked diffusion language models at each and every step commit the tokens at the positions where they are most confident; however, calibration remains audited over random masks. On teacher-forced traces from LLaDA-8B, Dream-7B and a 110M model of our own, we audit calibration conditional on this selection (C1). In the states a confidence decoder visits, the gap between a committed token's confidence and its accuracy grows with the number of candidates it beat in all seven model and task arms (Holm-corrected, held-out documents), and on the tokens it commits while 32–63 positions are still masked, confidence exceeds accuracy by 1.8 to 4.4 times the random-mask calibration error (descriptive). On splitting this gap, we isolate a field term, the part that is a winner's curse: in those states it grows for both instruct models on both tasks, though less with an additive calibrator, and on MBPP a shared-error term grows more; for the base models it is near the estimator's bias. C2 computes the dependency and estimation errors of committing several tokens at once by using a JSON probe with a known joint distribution, showing that dependency dominates only for strongly coupled tokens committed several at a time, and on real text a weak proxy for it is near zero below the all-mask state. C3 commits every position whose confidence after recalibration is at least $1 - \alpha$, with no extra forward pass. Finally, in an exploratory matched-speed comparison on GSM8K it beats decoders that commit a fixed number of tokens per step by 5 to 8 points, but not a raw confidence threshold: recalibration gives the cutoff a meaning, not extra accuracy.

## 1 Introduction

A masked diffusion model writes by unmasking: at each forward pass the decoder commits tokens at the positions it is most confident about. LLaDA commits $k$ of them, which is also an option in Dream, and decoding rules such as Fast-dLLM commit every position above a confidence threshold (Nie et al., 2025; Ye et al., 2025; Chang et al., 2022; Wu et al., 2025). Confidence is taken as how sure the model is about a particular option, so it is read as the chance the committed tokens are right. However, this confidence is also subject to noise, and the largest of many such estimates tends to be one whose error pushed it up. This is the winner's curse (Andrews et al., 2024), and a commit out of many masked positions is such a maximum.

Calibration of diffusion LMs, where measured, is over whole answers (Yadav et al., 2026; Jiang and Park, 2026) or per committed token without conditioning on the selection (Asaria et al., 2026). Nor is the bias automatic: selection does not bias a correct posterior (Dawid, 1994; Efron, 2011). We show that the gap between confidence and accuracy on committed tokens splits exactly, when positions' errors are independent given the state, into ordinary miscalibration and a *field term*, nonzero only if accuracy at fixed confidence depends on context (Proposition 2, §2). Only the field term is a winner's curse, and only an audit at the decoder's selection can see it.

Three contributions follow.

- **C1, a selection-conditional calibration audit** (§4). On teacher-forced traces from LLaDA-8B, Dream-7B and a 110M-parameter model that we trained, we measure the committed-token gap and its winner's-curse field term against how many candidates a token beat, with Holm-corrected tests on held-out documents. In the states a confidence decoder visits, the gap grows with field size for every model, and the field term grows for both instruct models on both tasks. For the base and 110M models the field term is near the bias of the estimator. At equal confidence, tokens are right less often in those

states than in random masks of the same size, in every model (exploratory).

- **C2, an instrument for the two errors of parallel commitment** (§5). Committing several tokens in one step incurs a dependency error (A) and an estimation error (B) (Eq. 1), computed exactly on a JSON probe with known joint distribution. (A) overtakes (B) only for strongly coupled slots ($\rho \geq 0.5$) and several tokens per step, and confidence ranking raises (A), not (B); the model's partial reading of $\rho$ also moves (B). On real text a weak top-token proxy for (A) is near zero below the all-mask state.
- **C3, a calibrated commitment policy** (§6). C3a commits as many positions as have recalibrated confidence at least $1 - \alpha$, in confidence order, with no extra forward pass. At matched speed (LLaDA-8B-Instruct, GSM8K) it beats top-$k$, margin and PC-Sampler but not entropy-bounded decoding or the Fast-dLLM threshold it recalibrates, which it trails at 8 tokens per step (exploratory).

## 2 Setup and theory

**Masked diffusion decoding.** A masked diffusion LM (Austin et al., 2021a; Sahoo et al., 2024; Nie et al., 2025; Ye et al., 2025) writes by unmasking: in a *state* $s$ with masked set $M = M_s$, one forward pass gives at each $i \in M$ a distribution $p_{\theta,i}$ given the observed tokens $x_{\text{obs}}$, with top-1 token $v_i$, *confidence* $c_i = p_{\theta,i}(v_i)$ and log-odds $\ell_i = \log \frac{c_i}{1-c_i}$, so $c_i = \sigma(\ell_i) = (1 + e^{-\ell_i})^{-1}$. The decoder *commits* positions for good: the $k$ most confident per step (Chang et al., 2022) or all above a cutoff (Wu et al., 2025). Committing $S \subseteq M$ in one step draws its tokens from the product of their marginals; with $p^*$ the true joint of $x_S$ given $x_{\text{obs}}$ and $p_i^*$ its marginals,

$$\mathrm{KL}\big(p^* \,\|\, \textstyle\prod_{i\in S} p_{\theta,i}\big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})}_{\text{(B) estimation}}, \quad (1)$$

where $\mathrm{TC}(x_S \mid x_{\text{obs}}) = \sum_{i\in S} H(p_i^*) - H(p^*)$ is the total correlation, zero iff they are conditionally independent. (A) is what no product of marginals can represent, (B) the marginals' error (cf. Ben-Hamu et al., 2025).

**The committed-token gap.** Under teacher forcing (§3) correctness $a_i = \mathbf{1}\{v_i = x_{0,i}\}$ against the true document $x_0$ is observed. A commit is a selection, so we model a field of $m$ candidates as a random pool $P \subseteq M$, won by $i = \arg\max_{j\in P} \ell_j$ (ties at random). With $\mathbb{E}_m$ averaging over states, size-$m$ pools and winners, the *committed-token gap* is

$$\Delta(m) = \mathbb{E}_m\,[\,c - a\,], \quad (2)$$

confidence minus accuracy, so positive means overconfident. $\Delta(1)$ is the ordinary signed calibration error of a masked position; larger $m$ stands for a larger field, as at early steps and in long outputs, and random pools vary $m$ with the state held fixed.

**The winner's curse.** If the winner's curse acts on decoding, $\Delta(m)$ grows with $m$. Let $\pi(\ell) = \Pr(a = 1 \mid \ell)$ be the calibration curve over all masked positions (the $m = 1$ population) and $\pi(\ell, s) = \Pr(a = 1 \mid \ell, s)$ its context-aware version.

**Proposition 1 (Winning adds nothing)** *Fix $s$. Suppose that, given $s$, each pair $(\ell_i, a_i)$ is independent of the other positions' scores $(\ell_j)_{j\neq i}$, and that the pool and the tie-breaks are drawn independently of all $(\ell_j, a_j)_{j\in M}$. Then the winner $i$ of $P$ satisfies*

$$\Pr\big(a_i{=}1 \;\big|\; \ell_i,\, s,\, i = \arg\max_{j\in P} \ell_j\big) = \pi(\ell_i, s).$$

Winning depends only on the other scores, which say nothing about $a_i$, so selection does not bias a correct posterior (Dawid, 1994; Efron, 2011); it acts only through context:

**Proposition 2 (Exact split of the gap)** *Under the conditions of Proposition 1,*

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\textit{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\textit{field term}}, \quad (3)$$

*and the field term is exactly zero at $m = 1$.*

Proofs are in Appendix A.3. Own miscalibration, reweighted towards where winners sit, can make $\Delta(m)$ grow with $m$ with no winner's curse; the *field term*, nonzero only if accuracy at fixed confidence depends on context (it requires a nonzero grouping loss; Kull and Flach, 2015;

| Model | Text | Prec. | Docs |
|---|---|---|---|
| LLaDA-8B-Base | FineWeb-Edu | fp16 | 1,979 |
| Dream-7B-Base | FineWeb-Edu | fp16 | 1,976 |
| Our 110M MDLM | FineWeb-Edu | fp32 | 1,979 |
| LLaDA-8B-Instruct | GSM8K | fp16 | 1,319 |
| LLaDA-8B-Instruct | MBPP | fp16 | 499 |
| Dream-7B-Instruct | GSM8K | fp16 | 1,319 |
| Dream-7B-Instruct | MBPP | fp16 | 499 |

Table 1: C1 trace runs: the seven primary arms. Also run: the time-split set (500 documents for each base model and the 110M model, same precision) and LLaDA-8B-Base in 4-bit on 198 FineWeb-Edu documents. In all, 0.49B position rows, 208 GPU-hours.

Perez-Lebel et al., 2023), is the winner's curse in a testable form. If positions of a state share errors, Proposition 1 fails and (3) gains a third term, $\mathbb{E}_m[\pi(\ell, s) - a]$ (§3); it also gains one if $\pi(\ell, s)$ pools a state's positions, as our calibrators do, and they differ in accuracy at equal $\ell$ (Appendix A.3).

## 3 Measurement instrument

### 3.1 Models and data

**Models.** We audit LLaDA-8B (Nie et al., 2025) and Dream-7B (Ye et al., 2025), Base and Instruct, in fp16. As neither discloses its pretraining data, we also trained a 110M MDLM (Sahoo et al., 2024) on data we control: a bidirectional transformer (12 layers, width 768, 12 heads, GPT-2 BPE plus a mask token, length 256) trained on 10.0B tokens (169,055 steps) of FineWeb-Edu sample-10BT (Penedo et al., 2024). Its training stream excludes the calibration and reserve documents and matches no time-split document.

**Data.** We evaluate models on sampled FineWeb-Edu documents, the GSM8K (Cobbe et al., 2021) and MBPP (Austin et al., 2021b) test sets, and a post-release Wikipedia set for robustness, with a 50/50 fit-and-report split; overly long documents are left out, and every C1 number comes from the report half. A document is contaminated if a 7–8B Base model, given its first 50 tokens, reproduces a run of $\geq$ 32 of the next 50 (2/1,979 LLaDA, 1/1,976 Dream); FineWeb endpoints are also reported without them.

### 3.2 Traces

Each measurement is one forward pass on a true document with some positions masked, recording at each masked position $v_i$, $c_i$, $\ell_i$, $a_i$ (§2), the top-20 and whether it was committed. A document has three kinds of states: *uniform*, each position masked independently at rate $t \in \{0.1, 0.2, \ldots, 0.9\}$, 2 masks per rate; *decoder*, every state of a teacher-forced confidence top-$k$ trajectory, $k \in \{2, 4, 8\}$, which starts fully masked, commits the $k$ most confident positions, fills them with the *true* tokens and repeats (the states a decoder visits if always right); and *matched uniform*, a uniform mask of each decoder state's size, so only the pattern differs: 466 passes per 256-token document. In every stored state with at least 32 masked positions we draw 20 random pools of each size $m \in \{1, 2, 4, 8, 16, 32\}$, rank them by $\ell$ with random tie-breaks and keep the top four ranks; $\Delta(1)$ is exact over all masked positions of those states. Answers under 32 tokens give none: 8/659 GSM8K and 65/250 (LLaDA), 74/250 (Dream) MBPP report documents.

### 3.3 Estimands

**Calibrators.** Both are fitted on the fit half, per kind of state. $\hat{\pi}(\ell)$ is isotonic in $\ell$. $\hat{\pi}(\ell, s)$ is logistic in a restricted cubic spline of $\ell$ plus the state's field features (median $\ell$ of its masked positions, mask rate, $\log|M|$, decoding step, not in uniform states) and their products with $\ell$, so the slope varies too; an additive form, without these products, is a sensitivity check. Documents and their states weigh equally (pool weights, Appendix A.2). The calibrators are fitted on every state but the averages use only states with at least 32 masked positions, so the estimated field term need not be zero at $m = 1$: E2, a contrast, removes only an offset constant in $\ell$.

**E1, E2 and the native gap.** E1 is $\Delta(m)$ (2), with primary contrast $\Delta(32) - \Delta(1)$, beside 15-bin equal-mass ECE over all masked positions. E2 is the field term $\mathbb{E}_m[\hat{\pi}(\ell) - \hat{\pi}(\ell, s)]$ of (3), with primary contrast field(32)$-$field(1). On the same winners, own miscalibration $\mathbb{E}_m[c - \hat{\pi}(\ell)]$, the field term and $\mathbb{E}_m[\hat{\pi}(\ell, s) - a]$ sum to $\Delta$ exactly; the last, zero under Proposition 1, is a specification check, as is the intra-class correlation of residuals $a - \hat{\pi}(\ell, s)$ within states. The *native* gap is $c - a$ over the tokens the decoder commits, grouped by $|M_s|$ at commit time.

**Rank diagnostics.** Eq. 4 is on pool winners, Eq. 5 on the top four ranks, each pooled over pool sizes, one unweighted row per pool member;

$\eta_i = \alpha\,[+\alpha_s] + f(\ell_i)$:

$$\Pr(a_i{=}1) = \sigma\big(\eta_i + \beta_z z_{r,m}\big), \quad (4)$$

$$\Pr(a_i{=}1) = \sigma\big(\eta_i + \textstyle\sum_{k=2}^{4}\gamma_k \mathbf{1}\{r{=}k\}\big), \quad (5)$$

where $z_{r,m}$ is the expected $r$-th largest of $m$ standard normals ($r{=}1$ for winners) and $f$ a restricted cubic spline with 9 knots, fitted on the fit half and carried over as a fixed offset. Each is fitted without and with state fixed effects $\alpha_s$ (ridge-penalised, so separated states stay finite), which absorb the field term because it lives between states: $\beta_z < 0$ without them and $\approx$0 with them points to a field effect, $\beta_z \neq 0$ with them to shared errors; the reading is approximate, as a calibration slope that varies by state can also load onto $z_{r,m}$ (§3.5), so $\beta_z$ is a diagnostic. E3 (decoder against uniform states at equal confidence and mask rate) and the controls are defined in Appendix A.2.

### 3.4 Inference

**Bootstrap.** The document is the unit. Intervals are 95% percentile intervals over 10,000 document resamples, one matrix per run shared across pool sizes, estimands and arms, so contrasts come from differenced draws. Eqs. 4–5 are refit per resample and use 1,000 (knot runs: 200). Calibrators are not refitted: E2 intervals carry only the report half's spread in $(\ell, s)$, not the calibrators' estimation error, so they understate E2's uncertainty.

**Declaring effects, multiple testing.** An effect is declared only if its interval excludes 0 *and* its magnitude exceeds twice the non-determinism floor, the largest change in $\Delta(m)$ over five repeats of 20 reserve documents per model family: 0, as repeats were bitwise identical. Holm's correction (two-sided bootstrap $p$-values) runs within families: the E1 contrasts and the E2 contrasts, each over the seven arms of Table 1 in uniform and $k{=}4$ decoder states (14 tests each), and $\gamma_2, \gamma_3, \gamma_4$ within each arm and kind of state.

### 3.5 Validation

**Known truth.** We ran the pipeline on simulated traces (code archive) in four worlds: calibrated everywhere; miscalibrated alike in every context; miscalibrated by context (the winner's curse of Proposition 2); and errors shared within a state (violating Proposition 1). The E1 contrast was zero only in the first, and $\beta_z$ without fixed effects strongly negative only in the third. Three findings shape §4: the field-term estimator reads high (at $m = 32$, 0.001 and 0.002 where the truth is zero; above it otherwise); fixed effects do not fully clean $\beta_z$ there; and shared errors give a *negative* field term with a *positive* $\beta_z$ without fixed effects. Too few knots let curvature load onto $z_{r,m}$; we use 9.

## 4 Selection and confidence (C1)

C1 asks whether a committed token's confidence overstates its accuracy more when it wins a larger pool, and how much of that is the field term of Proposition 2.

**Arms and claims.** We claim an E1 or E2 effect only if it is declared *and* Holm rejects it at $\alpha = 0.05$ in its family: the seven arms in uniform and $k{=}4$ decoder states (§3.4; a missing cell counts as $p = 1$). Everything else is exploratory, with intervals and no claims: decoder states at $k \in \{2, 8\}$, matched uniform states, $\beta_z$, $\gamma_k$, E3, the native gap, the controls and the robustness runs.

### 4.1 Does the gap grow with the field?

Of the 14 E1 tests in Table 2, Holm rejects 12 and 14 are declared; 12 meet both and are claims. For E2 the counts are 14, 14 and 14. The signs differ, so we do not pool. E1 is positive and claimed in every $k{=}4$ decoder-state cell and in uniform states of Dream-7B-Instruct on both tasks: there a token that wins a field of 32 is more overconfident than a random masked position. It is negative and claimed in uniform states of both base models and of LLaDA-8B-Instruct on GSM8K, where the gap shrinks with the field. The two cells without a claim, uniform states of LLaDA-8B-Instruct on MBPP and of the 110M model, are both declared but not claimed (both $p_{\text{Holm}} = 0.0524$).

In decoder states E1 turns positive for both base models and LLaDA-8B-Instruct on GSM8K, and is claimed there for LLaDA-8B-Instruct on MBPP and the 110M model. Uniform states matched to the decoder's mask counts (exploratory) keep the base models' negative sign, so the change comes from which positions stay masked.

**How much is the field term?** E2 meets the claim rule in all 14 cells and is positive in 12. This holds only given the fitted calibrators, since its intervals leave out their error (§3.4). The baseline field(1) is near 0 when calibrators are fitted on averaged states alone (§3.3). It is at most 0.005, except in instruct decoder states (0.011 to 0.018), where E2 may also hold an offset that varies with

| Arm | Masks | E1 | Own | E2 | Third | $\beta_z$ | $\beta_z^{\mathrm{FE}}$ |
|---|---|---|---|---|---|---|---|
| LLaDA Base | uniform | $-0.003^{*,\dagger}$ [−0.004, −0.002] | −0.004 | $0.0010^{*,\dagger}$ [0.0009, 0.0010] | 0.000 | $-0.03^{\ddagger}$ | $-0.01^{\ddagger}$ |
| | dec. $k$=4 | $0.006^{*,\dagger}$ [0.005, 0.008] | 0.010 | $0.0036^{*,\dagger}$ [0.0035, 0.0037] | −0.008 | $0.01^{\ddagger}$ | 0.01 |
| Dream Base | uniform | $-0.002^{*,\dagger}$ [−0.004, −0.001] | −0.004 | $0.0011^{*,\dagger}$ [0.0011, 0.0012] | 0.001 | $-0.03^{\ddagger}$ | 0.00 |
| | dec. $k$=4 | $0.015^{*,\dagger}$ [0.013, 0.017] | 0.010 | $0.0029^{*,\dagger}$ [0.0028, 0.0029] | 0.002 | $-0.02^{\ddagger}$ | $-0.02^{\ddagger}$ |
| LLaDA-I GSM8K | uniform | $-0.022^{*,\dagger}$ [−0.025, −0.018] | −0.027 | $-0.003^{*,\dagger}$ [−0.004, −0.003] | 0.009 | $-0.16^{\ddagger}$ | −0.02 |
| | dec. $k$=4 | $0.024^{*,\dagger}$ [0.016, 0.032] | −0.004 | $0.021^{*,\dagger}$ [0.020, 0.023] | 0.006 | $-0.10^{\ddagger}$ | −0.02 |
| LLaDA-I MBPP | uniform | $0.025^{\dagger}$ [0.003, 0.051] | −0.041 | $0.013^{*,\dagger}$ [0.010, 0.016] | 0.054 | $-0.56^{\ddagger}$ | $-0.23^{\ddagger}$ |
| | dec. $k$=4 | $0.156^{*,\dagger}$ [0.126, 0.189] | 0.032 | $0.036^{*,\dagger}$ [0.031, 0.041] | 0.088 | $-0.04^{\ddagger}$ | 0.01 |
| Dream-I GSM8K | uniform | $0.061^{*,\dagger}$ [0.052, 0.070] | −0.019 | $0.021^{*,\dagger}$ [0.020, 0.023] | 0.059 | $-0.42^{\ddagger}$ | −0.03 |
| | dec. $k$=4 | $0.072^{*,\dagger}$ [0.064, 0.081] | 0.025 | $0.038^{*,\dagger}$ [0.036, 0.039] | 0.009 | $-0.05^{\ddagger}$ | $0.02^{\ddagger}$ |
| Dream-I MBPP | uniform | $0.257^{*,\dagger}$ [0.227, 0.290] | −0.018 | $0.176^{*,\dagger}$ [0.159, 0.194] | 0.100 | $-0.76^{\ddagger}$ | −0.07 |
| | dec. $k$=4 | $0.186^{*,\dagger}$ [0.151, 0.224] | 0.022 | $0.076^{*,\dagger}$ [0.066, 0.086] | 0.089 | $-0.19^{\ddagger}$ | 0.01 |
| MDLM-110M | uniform | $0.00152^{\dagger}$ [0.00003, 0.00301] | 0.000 | $0.0019^{*,\dagger}$ [0.0018, 0.0019] | 0.000 | $-0.01^{\ddagger}$ | 0.00 |
| | dec. $k$=4 | $0.009^{*,\dagger}$ [0.007, 0.011] | 0.013 | $-0.0023^{*,\dagger}$ [−0.0023, −0.0022] | −0.002 | $0.02^{\ddagger}$ | 0.01 |

Table 2: C1 primary endpoints. E1 = $\Delta(32) - \Delta(1)$ and E2 = field(32)−field(1), with 95% intervals (10,000 document resamples, calibrators held fixed for E2, §3.4; $\beta_z$: 1,000). Own, Third: the same contrast of own miscalibration and of $\mathbb{E}_m[\hat{\pi}(\ell, s) - a]$ (§3.3); Own + E2 + Third = E1 up to rounding. $^{\dagger}$declared; $^{*}$claimed (E2: given the fitted calibrators). $\beta_z$, $\beta_z^{\mathrm{FE}}$: Eq. 4 without and with state fixed effects ($^{\ddagger}$interval excludes 0).

$\ell$. In the base and 110M models every |E2| is below 0.005, as small as the estimator gave on simulated traces with no field term (§3.5), so we do not read a winner's curse there. In seven of the eight instruct cells E2 runs from 0.013 to 0.176 (Dream-7B-Instruct, MBPP, uniform states; E1 is 0.257). There the field term grows with $m$, as a winner's curse predicts: at equal confidence, winners of large fields come from contexts where tokens are right less often. The eighth (LLaDA-8B-Instruct, GSM8K, uniform) is negative. A spline-logistic $\hat{\pi}(\ell)$ of the same form as $\hat{\pi}(\ell, s)$ (Table 4, no control) keeps E2's sign in every cell and its size within 0.003. The exception is uniform states of Dream-7B-Instruct (GSM8K 0.014, MBPP 0.098), where the model form, not context, carries part of E2. Positions in a state share errors in every arm (residual ICC within states is above 0: at most 0.006 in base and 110M models, 0.036 to 0.056 in instruct models), so the split has a third term (Table 2, Third). It is near 0 in base and 110M models, except LLaDA-8B-Base decoder states. It reaches 0.054 to 0.100 on MBPP and in Dream-7B-Instruct uniform states on GSM8K, and is larger than E2 in four of these five cells. In those cells winners are right less often than $\hat{\pi}(\ell, s)$ predicts, so selection works more within states (shared errors, or positions that differ at equal $\ell$) than through the field term. E2 and $\beta_z$ should be read with this in mind.

### 4.2 The selection diagnostic $\beta_z$

$\beta_z$ (Eq. 4) asks whether winning a larger field predicts correctness beyond the winner's own log-odds; we read it beside the sign of E2 (§3.5). Without fixed effects it excludes 0 in all 14 cells (Table 2) and is negative in 12, and with them its interval holds 0 in 8 of those 12: the pattern of a field effect; E2 is positive in all of them but uniform states of LLaDA-8B-Instruct on GSM8K. Holm, run within each arm and kind of state, rejects 24 of the 42 tests of $\gamma_2, \gamma_3, \gamma_4$ (Eq. 5), in 8 of the 14 rows, all positive and all in the instruct models and Dream-7B-Base decoder states: within a state, at equal confidence, lower-ranked pool members are right more often than the winner, which Proposition 1 excludes. Like the ICC, we read this as a sign of shared errors (or of a calibration slope that varies by state); none is a claim.

### 4.3 Decoder states

E3 compares accuracy at equal confidence and mask rate in decoder and matched uniform states (Appendix A.2); the decoder is teacher-forced, so a difference comes from which positions remain masked, not from its own mistakes. It is positive, its interval excluding 0, in every arm and at every $k \in \{2, 4, 8\}$ (sup-$t$ $p \leq 0.002$; exploratory): tokens are right less often in the states a decoder visits, by 0.001 to 0.006 in the base and 110M models and 0.03 to 0.06 in the instruct models, so C3a's calibrator is fitted on decoder states (§6). The de-

coder's own commits are more overconfident still: with 32–63 positions masked, $c - a$ on the $k$=4 decoder's commits is 1.8 to 4.4 times its arm's random-mask ECE in every arm, its interval above it (descriptive; Appendix A.2).

### 4.4 Controls and robustness

Position, token type and a temperature per mask-rate bin leave decoder-state E1 with its sign in every arm, but not LLaDA-8B-Base's small uniform-state E1, which token type takes to 0; a 4-bit check does not rule out an effect of precision (Appendix A.2). Five checks repeat the contrasts outside the Holm families (code archive). *Contamination:* without the 2 LLaDA and 1 Dream documents the screen flagged (§3.1), both base models' E1 equal their main values. *Time split:* on unseen Wikipedia articles (a quarter of the documents) the small negative uniform-state E1 of the base models is not reproduced (LLaDA-8B-Base: $-0.001$ $[-0.003, 0.002]$): seen text, the change of domain, or too small to detect here; the 110M model's grows (0.012); it saw neither set, so domain, not contamination, moves it. Decoder-state E1 stays positive, its interval excluding 0, for all three. *Knots:* E2 keeps its sign and declared flag with 5 and 13 knots in 14 of 14 cells. *Calibrator form:* with the additive calibrator (§3.3) it does in 12 of 14 cells. The two exceptions, claimed cells whose E2 we therefore do not rely on, are uniform states of LLaDA-8B-Instruct on MBPP, where E2 falls from 0.013 to 0.000 (interval holding 0), and 110M decoder states, where it goes from $-0.002$ to $0.001$. The other six positive instruct-model E2s shrink to 0.005–0.098, two of them (Dream-7B-Instruct on GSM8K, uniform; LLaDA-8B-Instruct on MBPP, decoder) to 0.005 and 0.007, close to the base models' E2. *Independent estimator:* an exact all-subsets estimator with its own calibrator (after Dorigo 2026) reproduces E1 in all four base cells and matches E2 in sign and size (at most the $0.003$ it reads with no field term), except in LLaDA decoder states ($-0.0004$ vs $0.004$).

## 5 Two sources of error (C2)

Parallel decoders target the dependency term (A) of Eq. 1 and take the marginals as accurate (§7).

### 5.1 A JSON probe with known $p^*$

The probe fills the simple fields of 200 SchemaStore schemas (SchemaStore contributors, 2026), 6–16 per schema, with single-token values; we call them slots. Each slot has 2–10 allowed values and a fixed Dirichlet(1) marginal; slots are paired: with probability $\rho$, stated in a marker at the start of the instance, the follower is a fixed map of its leader's value (spread evenly over its own values), otherwise an independent draw, so $p^*$ is known exactly. We fine-tune our 110M MDLM (§3) on 200k instances and evaluate 1,000 (200 per $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$); the main model never saw the marker masked (Appendix A.1). Decoding starts with every value masked; each step picks $k \in \{1, 2, 4, 8\}$ slots (the last step, the rest), the most confident or at random, scores (A) and (B) ($p_{\theta,i}$: the model's probability of each allowed value, unnormalised, so mass off the list counts against (B)), and fills them with true values. Terms are summed over steps and divided by the number of slots; intervals use 10,000 instance resamples.

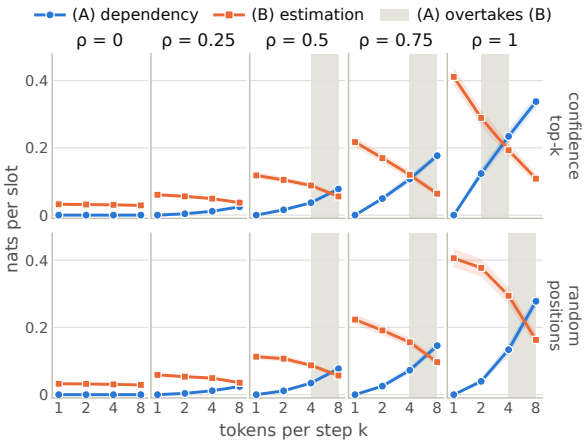



Figure 1: (A) and (B) per slot (main model), with 95% intervals, against $k$. Columns: $\rho$; rows: selection rule. Shaded: the $k$ where (A) exceeds (B).

### 5.2 (A) overtakes (B) at high $\rho$ and $k$

(A) is zero at $k = 1$ and at $\rho = 0$ and grows with both (Figure 1). For $\rho \geq 0.5$ it overtakes (B) at $k = 8$, or at $\rho = 1$ under confidence top-$k$ already at $k = 4$; each crossover is also where the 95% interval of (A)−(B) clears zero. For $\rho \leq 0.25$ (B) dominates up to $k = 8$. At $\rho = 1$, $k = 8$ (confidence), (A) is 0.338 [0.320, 0.355] nats per slot and (B) 0.109 [0.095, 0.123]: 76% of the total.

**Confidence selection raises (A), not (B).** Where (B) is higher under confidence ranking than at random, the intervals overlap. (A) rises instead ($\rho = 1$, $k = 2$: 0.124 [0.108, 0.140] against 0.040 [0.031, 0.049]): the ranking commits leaders with their followers more often than

chance, bringing the crossover forward.

**The terms trade off.** $\rho$ was meant to move only (A), but (B) rises too: at $k = 1$, from 0.032 at $\rho = 0$ to 0.411 at $\rho = 1$. Decoded after its leader, a follower should follow from it, but the model reads the marker only in part (Appendix A.1) and stays broad: error in (B). Decoded in the same step, the true marginal is broad too, and the missing coupling counts as (A). At $\rho = 1$, from $k = 1$ to 8, (B) falls to 0.109 while (A) rises from 0 to 0.338, so (B)'s fall moves the crossovers too. (A) exceeds the $\rho = 0$ (B), 0.032, from $k = 4$ at $\rho = 0.5$ and $k = 2$ at $\rho = 1$. Without fine-tuning, (B) dominates everywhere ((A) is $\leq 4\%$ of the total).

### 5.3 Real text: proxies checked on the probe

The exact values hold for one synthetic distribution and one 110M model; on real text $p^*$ is unknown, so we use proxies. For (A), parallel–sequential disagreement: the fraction of the $k$ most confident masked positions whose token changes if committed one at a time (earlier ranks filled with the model's own tokens) instead of in one pass. For (B), the gap $\Delta = c - a$ and the cross-entropy (CE) of the true token, at the same positions. We run the 110M model and LLaDA-8B-Base on 300 fit-half FineWeb-Edu documents, one state at each mask rate $0.1, 0.2, \ldots, 1$ (Appendix A.1), and the proxy on the probe.

**The proxy tracks (A) weakly.** Over the 15 confidence-rule cells with $k \geq 2$, its Spearman correlation with the exact (A) is 0.49 [0.28, 0.63]. It rises with $k$, barely with $\rho$: at $k = 8$, 0.028 at $\rho = 0$, where (A) is zero, and 0.049 at $\rho = 1$.

**On real text the (A) proxy is near zero.** Over mask rates 0.1–0.9, disagreement at $k = 8$ is 0.0021 for the 110M model and 0.0022 for LLaDA, and at most 0.0003 at $k \leq 4$; at rate 1 (all-`[MASK]` input), where free generation starts, it is 0.6250 for the 110M model and 0 for LLaDA at $k = 8$. CE grows from 0.08 nats at $k = 1$ to 0.40 at $k = 8$ (LLaDA: 0.08 to 0.23), though it includes the text's own entropy and lower ranks are harder positions. $\Delta$ cannot be told from zero at $k = 1$ and is small but positive at $k = 8$: $+0.007$ [0.004, 0.010] and, for LLaDA, $+0.004$ [0.002, 0.007].

So below the all-mask state, committing 2–8 tokens at once rarely changes the models' top token. The proxy sees only dependence strong enough to flip a top token and barely moves with $\rho$ on the probe, so this does not show that (A) is small; real text gives a direction, not a split in nats.

## 6 A recalibrated commit rule (C3)

**The rule.** A top-$k$ decoder fixes the number of tokens per step in advance; C3a lets calibrated confidence set it. At each step it computes $\hat{c}_i = \hat{\pi}(\ell_i, s)$ with the context-aware calibrator of §3.3 on three field features (median $\ell$, mask rate, $\log |M|$, over the whole generated region, not the active block), and commits the $n$ masked positions of the active block with the highest $\ell$, where $n$ is the number with $\hat{c}_i \geq 1 - \alpha$, or one if none qualifies. If $\hat{c}$ is non-decreasing in $\ell$ across a step's positions, these are exactly the positions with $\hat{c}_i \geq 1 - \alpha$, so every committed token except a forced single commit has calibrated error probability $1 - \hat{c}_i \leq \alpha$. The $\ell\times$feature terms do not guarantee this, but it holds in every fit state and every checked report-half state (free generation was not checked). $\alpha$ is the one speed knob. C3a changes *how many* tokens are committed, not *which*, and needs no extra forward pass.

**The calibrator.** One calibrator per task is fitted on teacher-forced decoder states ($k \in \{2, 4, 8\}$) of the fit-half LLaDA-8B-Instruct traces (660 GSM8K and 249 MBPP documents) and frozen. On held-out report-half states its ECE falls to 0.004 from raw $\sigma(\ell)$'s 0.060 on GSM8K and to 0.005 from 0.040 on MBPP. There, tokens C3a commits above the cutoff are wrong less often than $\alpha$ at every $\alpha$ in $[0.01, 0.6]$ (0.07–0.09 at $\alpha = 0.25$); forced single commits (35–50% of states) are wrong about half the time. It is then applied to free generation, which starts from 256 masks; its fit states lie inside a reference answer. Of C3a's steps, 0.017 (GSM8K) and 0.034 (MBPP) have features outside the fitted range.

**Matched speed.** LLaDA-8B-Instruct generates freely on the report halves of GSM8K (659 problems) and MBPP (121): zero-shot chat prompt, 256 tokens in blocks of 32, greedy, one seed, fp16. C3a is compared with confidence top-$k$, Fast-dLLM's threshold (Wu et al., 2025), margin (Kim et al., 2025a), entropy-bounded (Ben-Hamu et al., 2025) and full PC-Sampler (Huang et al., 2025); our KLASS (Kim et al., 2025b) run had implementation errors (Appendix A.4). Each decoder has one speed knob, set per task on fit-half problems, by speed alone, to 2, 4 and 8 tokens per

| | GSM8K ($N = 659$) | | | MBPP ($N = 121$) | | |
|---|---|---|---|---|---|---|
| Decoder | tok/step | Acc. [95% CI] | C3a − decoder [95% CI] | tok/step | Acc. [95% CI] | C3a − decoder [95% CI] |
| Top-$k$ | 4.00 | 0.730 [0.695, 0.762] | 0.059 [0.026, 0.093] | 4.00‡ | 0.207 [0.144, 0.287] | — |
| Threshold | 4.03 | 0.791 [0.758, 0.820] | −0.002 [−0.014, 0.012] | 3.73 | 0.355 [0.276, 0.444] | 0.000 [−0.033, 0.033] |
| Margin | 4.00 | 0.736 [0.701, 0.768] | 0.053 [0.020, 0.088] | 4.00‡ | 0.174 [0.116, 0.251] | — |
| Entropy | 4.28 | 0.794 [0.761, 0.823] | −0.005 [−0.035, 0.026] | 3.83 | 0.355 [0.276, 0.444] | 0.000 [−0.058, 0.058] |
| PC-Sampler | 4.00 | 0.707 [0.671, 0.741] | 0.082 [0.050, 0.112] | 4.00‡ | 0.157 [0.103, 0.232] | — |
| C3a | 4.04 | 0.789 [0.756, 0.819] | ref. | 3.61 | 0.355 [0.276, 0.444] | ref. |

Table 3: Decoders at matched speed: speed 2 (target 4 tokens per step), C3a at $\alpha = 0.271$ (GSM8K) and 0.322 (MBPP). Wilson intervals for accuracy, paired bootstrap for C3a − decoder (positive favours C3a). KLASS, run with implementation errors, is not shown (Appendix A.4). ‡Over 10% faster than C3a, so not paired.

step (speeds 1 to 3). Speed is aggregate tokens per step, $256N/\sum \mathrm{NFE}$ (NFE: forward passes) over a configuration's $N$ problems. Table 3 shows speed 2; Appendix A.4 has the sanity check and the speed curves. Each decoder is taken at its setting nearest to C3a's speed, within $\pm 10\%$ (else —), and paired with it on the shared problems (95% percentile bootstrap, 2,000 resamples). It is exploratory, without multiple-testing correction.

**Result.** At matched speed on GSM8K, C3a is indistinguishable in accuracy from the threshold and entropy-bounded decoders (C3a − threshold $-0.002$ $[-0.014, 0.012]$) and 0.053 to 0.082 more accurate than top-$k$, margin and PC-Sampler, all fixed-count (Table 3): the gain is in how many tokens are committed per step, which a raw threshold already achieves. Margin and PC-Sampler ran with a fixed count only, so their scores are not separated from that choice. On MBPP (the 121 of 250 problems all decoders finished before compute ran out; a later run of all 250, on two machines and so set in advance as a check only, agrees, code archive) it is again indistinguishable from both; fixed-count decoders, 11% faster, are unpaired at speed 2; at speeds 1 and 3 C3a is 0.017 to 0.083 more accurate (margin's intervals hold 0).

## 7 Related work

**Decoders.** LLaDA and Dream (Nie et al., 2025; Ye et al., 2025) scale masked diffusion (Sahoo et al., 2024) to 7–8B parameters and commit the most confident positions, as MaskGIT did (Chang et al., 2022); we audit that rule and propose no model. Other decoders replace top-1 probability (Kim et al., 2025a) with entropy, margin or stability (Ben-Hamu et al., 2025; Kim et al., 2025b), fix the cutoff (Wu et al., 2025), set it per block and step from a profiled run's confidences (Shen and Ro, 2025) or learn the policy (Jazbec et al., 2026); they evaluate speed and accuracy, not the calibration of the tokens they commit. C3a thresholds a probability of being right, fitted on labelled states.

**Calibration and the winner's curse.** Calibration is usually measured over all predictions (Guo et al., 2017). PC-Sampler (Huang et al., 2025) recalibrates the commit statistic for position and token frequency; we add its positional weight as a covariate. Keeping the largest of noisy estimates biases it upward (Andrews et al., 2024), more so as the pool grows (Dalal et al., 2026), but leaves a correct posterior unbiased (Dawid, 1994; Efron, 2011). Diffusion LMs are overconfident against autoregressive models on most tasks (Yadav et al., 2026), an entropy-based commit order lowers their sequence-level calibration error (Jiang and Park, 2026), and commit confidence tracks correctness on math but not on factual recall (Asaria et al., 2026); none conditions on how many candidates a committed token beat.

**Dependency between tokens.** Dependency-aware decoders (Kim et al., 2026; Ringel et al., 2026; Azangulov et al., 2026) and ParallelBench (Kang et al., 2026) target term (A) of Eq. 1 and take the marginals as accurate. The split is not ours: EB-Sampler uses the same two terms (Ben-Hamu et al., 2025), and Tang et al. (2026) note that sampler and model error stay entangled; C2 computes both in nats where $p^*$ is known.

## 8 Conclusion

Confidence-based decoders use one number both to choose which tokens to commit and as the chance that they are right; checked at the decoder's own selection, the first use biases the second. Dependency error dominates parallel commits only for strongly coupled tokens, and recalibrating the commit threshold gives the cutoff an error rate, not extra accuracy.

## Limitations

**Scope.** We evaluate two 7–8B families (fp16 traces) and one 110M MDLM across FineWeb-Edu, GSM8K and MBPP, with model and text type confounded. C2 covers one synthetic distribution and one 110M fine-tune, which reads the $\rho$ marker only in part, so (B) moves with $\rho$ and $k$ too; on real text it has only a top-token proxy for (A), and a divergence between parallel and sequential distributions was not computed.

**Design and estimators.** Since C1 is teacher-forced, its states are those of an always-right decoder, and a valid rewording counts as an error. The two-term split of the gap assumes positions' errors are independent given the state; they are not, and where the third term this adds is large (§4.1), E2 alone does not measure what selection does to the gap. The field term is only as context-aware as the state features of $\hat{\pi}(\ell, s)$: up to four, three in uniform states, where the step is constant. Its size also depends on how they enter: without products of $\ell$ and the features, the instruct models' E2 shrinks (§4.4), and on the population the calibrators are fitted on: we did not refit them on only the states the averages keep (§3.3). E2's intervals, and so its claims, leave out the calibrators' estimation error: a bootstrap that also refits them on resampled fit halves was not run (§3.4). Nor was a fit of $f$ jointly with the state effects in Eqs. 4–5, which would show how much a slope that varies by state moves $\beta_z$.

**Contamination.** The screen flags only verbatim recall, and the time-split set mixes exposure with domain. Only the 110M model isolates the domain part. Our claims are about change with field size (E1, E2), not levels; the levels we report (E3, the native gap) are descriptive.

**Exploratory analyses and C3.** Analyses outside the E1/E2 Holm families (decoder states at $k \in \{2, 8\}$, matched uniform states, $\beta_z$, $\gamma_k$, E3, the native gap, the controls, and the robustness runs of §4.4) carry no claims. C3a is fitted on teacher-forced states but applied to free-running generation, and compared on one model, against static cutoffs only (not dynamic ones, Shen and Ro 2025), with greedy decoding, one seed and two tasks. Its calibration and its monotonicity in $\ell$ are checked on held-out teacher-forced states, not in free generation, which has no reference token per position. There its features span the whole 256-token canvas, future blocks included, unlike the answer-only states it is fitted on; beyond the range check, the effect of this shift is unmeasured, and how often a step falls back to a forced single commit was not logged.

## Code and data

Code: https://github.com/abhirajratna/ANLP-Project
Models and traces: https://huggingface.co/ANLP-project

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# A Supplementary details

## A.1 C2: the main model and real-text details

**Main model.** One fine-tune masked the $\rho$ marker like any token (100M tokens); the main model never did, as at evaluation (300M; about 6 and 18 passes over the 200k instances). Neither moved the mapped value's probability by our bar of 0.5 when the marker is switched from $\rho = 1$ to 0 with only the leader observed (main 0.251, other 0.014), so we took the lower validation NELBO (main 0.1751, other 0.1892 nats per token; the main model also trained on more tokens; no checkpoint of it at 100M was kept). The other crosses over at $k = 8$ for every $\rho \geq 0.5$ under both rules.

**Real text.** The 300 documents are the fit-half ids up to 601 (600 for LLaDA) less those under 256 tokens; we keep at least 8 masked positions per state. Intervals resample documents (10,000 times).

## A.2 C1: E3, controls and committed tokens

**Weighting.** Averages are taken within a document, every state weighing the same, then across documents; documents without pools drop out. A state's 20 winners at each $m$ weigh equally, and its positions $1/|M_s|$ at $m = 1$. Calibrators weight positions by $1/(|M_s| \times$ states of that kind in their document); decoder states are fitted per $k$. Knots sit at evenly spaced quantiles of fit-half $\ell$, 5% to 95%. In Eqs. 4–5, $m = 1$ rows are the drawn singletons and $\alpha_s$ has a ridge of $10^{-3}$ on the unweighted log-likelihood.

**E3, trajectory shift.** On the positions of decoder states at each $k$ and of matched uniform states, mask rate in $[0.05, 0.95)$, we fit $\Pr(a{=}1) = \sigma(\alpha_b + f(\ell) + \mathit{dec} \cdot (g_0 + h(\ell)))$, $\alpha_b$ per mask-rate bin (0.1 wide, centred on $0.1, \ldots, 0.9$), $f$, $h$ joint splines on 9 shared knots. The endpoint $\mathbb{E}_{\mathrm{dec}}[\pi_{\mathrm{ref}}(\ell, b) - \pi_{\mathrm{dec}}(\ell, b)]$ is positive if decoder-state tokens are right less often at equal confidence and mask rate; a sup-$t$ test checks $(g_0, h)$ jointly.

**Controls.** *Position* (absolute position, distance to the nearest unmasked token, PC-Sampler's position weight; Huang et al., 2025) and *token type* (of the true token: no letter or digit, an NLTK stopword, or content) are covariates $x$: $\mathbb{E}_m[\hat{\pi}(\ell) - \hat{\pi}(\ell, x)]$ is removed from $\Delta(m)$, and E2 becomes $\mathbb{E}_m[\hat{\pi}(\ell, x) - \hat{\pi}(\ell, s, x)]$. *Temperature* fits $c = \sigma(\ell/T)$ per mask-rate bin on the fit half. Table 4 gives each control. For LLaDA-8B-Base in uniform states the change in E1 is below 0.0001 for position, 0.0029 [0.0027, 0.0032] for token type and 0.0005 [0.0004, 0.0005] for a temperature per mask-rate bin. In decoder states E1 keeps its sign under every control in every arm and moves by about 0.03 at most (temperature, LLaDA-8B-Instruct on MBPP: 0.156 to 0.125): the growth is not where winners sit, what kind of token they are, or an over- or underconfidence by mask rate. In uniform states the small base and 110M contrasts move by up to 0.004, so their sign is not robust. On the 4-bit run's report half (198 documents, calibrators from its fit half), NF4 in place of fp16 changes LLaDA-8B-Base's E1 by 0.0002 $[-0.0014, 0.0018]$ (uniform) and 0.0055 $[-0.0004, 0.0115]$ (decoder states): both intervals hold 0, but the latter is as large as fp16 E1 on these documents (0.0003), so a precision effect is not ruled out. Decoder-state E2 changes by $-0.0025$ $[-0.0030, -0.0021]$, of the size of the base models' E2.

**Committed tokens.** For tokens the $k{=}4$ decoder commits with 32–63 positions masked, $c - a$ is 0.079 [0.071, 0.086] for LLaDA-8B-Instruct on GSM8K, against an ECE of 0.033 over random masks; it is 0.013 to 0.018 for the base and 110M models and 0.079 to 0.176 for the instruct models. The mask count mixes field size with lost context, so this describes and does not test E1.

## A.3 Proofs of Propositions 1 and 2

Notation is as in §2, with $\ell_{-i} = (\ell_j)_{j \in M \setminus \{i\}}$, tie-break randomness $U$, and $W_i$ the event that $i \in P$ wins. Probabilities are over a superpopulation of $(\ell_j, a_j)$ given $s$, with $P$ uniform on the index set $M_s$; $\pi(\ell_i, s) = \Pr(a_i{=}1 \mid \ell_i, s)$ may depend on $i$ (suppressed). A calibrator pooled over a state's positions equals it only if they are exchangeable given $s$; otherwise the difference joins the third term.

**Proposition 1.** Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$. Given $s$, $(P, U)$ is independent of $(\ell_j, a_j)_{j \in M}$ and $(\ell_i, a_i)$ of $\ell_{-i}$, so the law of $(\ell_i, a_i, \ell_{-i}, P, U)$ given $s$ factorises as $p(P, U \mid s)\, p(\ell_i, a_i \mid s)\, p(\ell_{-i} \mid s)$, i.e. $(\ell_i, a_i) \perp Y \mid s$. By weak union $a_i \perp Y \mid Z$, so $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$ a.s. As $W_i$ is a function of $(Y, Z)$, conditioning on $(Y, Z)$ gives, for every bounded measurable $h$,

$$\mathbb{E}[a_i\, h(Z)\, \mathbf{1}_{W_i}] = \mathbb{E}[\pi(\ell_i, s)\, h(Z)\, \mathbf{1}_{W_i}], \qquad (6)$$

| Arm | Masks | E1 | +pos. | +type | +temp. | E2 | +pos. | +type | +temp. |
|---|---|---|---|---|---|---|---|---|---|
| LLaDA Base | uniform | $-0.003$ | $-0.003^{\ddagger}$ | $0.000^{\ddagger}$ | $-0.002^{\ddagger}$ | 0.001 | $0.001^{\ddagger}$ | $0.003^{\ddagger}$ | — |
| | dec. $k$=4 | 0.006 | $0.005^{\ddagger}$ | $0.008^{\ddagger}$ | $0.006^{\ddagger}$ | 0.003 | $-0.003^{\ddagger}$ | $0.008^{\ddagger}$ | — |
| Dream Base | uniform | $-0.002$ | $-0.002$ | $-0.001^{\ddagger}$ | $-0.002^{\ddagger}$ | 0.001 | $0.001^{\ddagger}$ | $0.002^{\ddagger}$ | — |
| | dec. $k$=4 | 0.015 | $0.015^{\ddagger}$ | $0.021^{\ddagger}$ | $0.016^{\ddagger}$ | 0.002 | $-0.001^{\ddagger}$ | $0.008^{\ddagger}$ | — |
| LLaDA-I GSM8K | uniform | $-0.022$ | $-0.019^{\ddagger}$ | $-0.022^{\ddagger}$ | $-0.014^{\ddagger}$ | $-0.003$ | $-0.001^{\ddagger}$ | $-0.004^{\ddagger}$ | — |
| | dec. $k$=4 | 0.024 | $0.022^{\ddagger}$ | $0.023^{\ddagger}$ | $0.028^{\ddagger}$ | 0.021 | $0.015^{\ddagger}$ | $0.022^{\ddagger}$ | — |
| LLaDA-I MBPP | uniform | 0.025 | 0.022 | $0.024^{\ddagger}$ | $0.014^{\ddagger}$ | 0.013 | $0.037^{\ddagger}$ | $0.012^{\ddagger}$ | — |
| | dec. $k$=4 | 0.156 | 0.153 | $0.153^{\ddagger}$ | $0.125^{\ddagger}$ | 0.039 | $0.069^{\ddagger}$ | $0.033^{\ddagger}$ | — |
| Dream-I GSM8K | uniform | 0.061 | 0.061 | $0.059^{\ddagger}$ | $0.075^{\ddagger}$ | 0.014 | $0.035^{\ddagger}$ | $0.012^{\ddagger}$ | — |
| | dec. $k$=4 | 0.072 | $0.070^{\ddagger}$ | $0.070^{\ddagger}$ | $0.059^{\ddagger}$ | 0.038 | $0.032^{\ddagger}$ | $0.039^{\ddagger}$ | — |
| Dream-I MBPP | uniform | 0.257 | $0.232^{\ddagger}$ | 0.254 | 0.260 | 0.098 | $0.120^{\ddagger}$ | $0.086^{\ddagger}$ | — |
| | dec. $k$=4 | 0.186 | $0.168^{\ddagger}$ | $0.183^{\ddagger}$ | $0.163^{\ddagger}$ | 0.077 | 0.078 | $0.075^{\ddagger}$ | — |
| MDLM-110M | uniform | 0.002 | $0.002^{\ddagger}$ | $0.006^{\ddagger}$ | $0.000^{\ddagger}$ | 0.002 | $0.002^{\ddagger}$ | $0.005^{\ddagger}$ | — |
| | dec. $k$=4 | 0.009 | $0.010^{\ddagger}$ | $0.011^{\ddagger}$ | $0.010^{\ddagger}$ | $-0.003$ | $-0.006^{\ddagger}$ | $0.003^{\ddagger}$ | — |

Table 4: Controls: E1 and E2 without a control and with position, token type, or a temperature per mask-rate bin (it rescales confidence only: no E2). E2 here uses logistic calibrators for both $\hat{\pi}(\ell)$ and $\hat{\pi}(\ell, s)$, so with and without differ only by the control; its "without" value differs from the isotonic E2 of Table 2. $^{\ddagger}$the paired with-minus-without difference has a 95% interval excluding 0. Exploratory.

which is the defining property of $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s. on $W_i$. □

**Proposition 2.** Let $\mathbb{E}_m$ weight states by any function of $s$, let $(\ell, a)$ be the winner's pair, and take $\pi(\ell)$ as the calibration curve of the $m = 1$ population under the same weights. Exactly one $W_i$ holds per pool, so $a = \sum_{i \in M} a_i \mathbf{1}_{W_i}$ and $\pi(\ell, s) = \sum_{i \in M} \pi(\ell_i, s) \mathbf{1}_{W_i}$. Summing (6) with $h \equiv 1$ over $i$ under the law of $\mathbb{E}_m$ (it holds given each $s$, so for any state weights) gives $\mathbb{E}_m[a] = \mathbb{E}_m[\pi(\ell, s)]$. Hence $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$, and adding and subtracting $\pi(\ell)$ gives (3). At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property, $\pi$ being the calibration curve of this very population, and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ as just shown, so the field term is zero. □

The estimator inherits this last step only if the calibrators are fitted on the averaged population; ours are not (§3.3), so E2 is a contrast.

### A.4 C3: sanity check, KLASS and speed curves (exploratory)

Before the comparison, the harness decodes 200 fit-half GSM8K problems at Fast-dLLM's settings (5-shot, 256 tokens, blocks of 32; fp16 on an RTX 2080 Ti). Top-$k$ at $k = 1$ scores 0.835 [0.777, 0.880] against a published 0.793, and the threshold rule at $\tau = 0.9$ scores 0.845 [0.788, 0.889] against 0.792 (Wu et al., 2025) (95% Wilson intervals): both published values lie inside, so the harness reproduces the two decoders it can be checked on, though our points

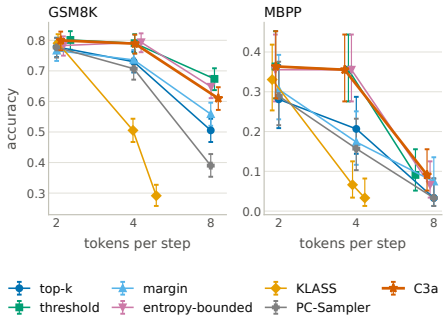



Figure 2: Accuracy against aggregate tokens per step, one panel per task (left GSM8K, right MBPP), one curve per decoder at its three speeds, with 95% Wilson intervals. Up and to the right is better.

are higher. At speed 3 (about 8 tokens per step; Figure 2) C3a trails the threshold rule on GSM8K (C3a − threshold $-0.064$ $[-0.106, -0.027]$); costs and the speed-1 and speed-3 rows are in the code archive.

**KLASS.** Ours differs from the reference (Kim et al., 2025b) in three ways, found after scoring: a KL history per 32-token block (two one-token steps at each block start), the reverse KL direction, and 32-token blocks, not 64. At its published thresholds it runs at about 2.6 tokens per step, as published; aiming at 4 and 8, the knob path, fixed in advance, lowers $\tau$ to 0.2–0.29, far below the published range. Its curve in Figure 2 describes that path, not KLASS as intended; Table 3 omits it.